

Data Warehousing with IBM Cognos

Speaking script for the 9-slide presentation. About 5 to 6 minutes at a calm pace.

Slide 1: Cover

Good morning everyone. My topic is data warehousing with IBM Cognos.

IBM's motto is one word: THINK. So I have built this talk the way IBM itself works, using three steps: Observe, Reflect and Make. First we observe a real problem. Then we reflect on a solution. Then we make it, using IBM's own tools.

Slide 2: The problem (Observe)

Let us observe the problem. A large hospital cannot see, on any given day, how long its patients are waiting or how many beds are free.

There are three reasons. First, the data is scattered: registration, lab, pharmacy and billing each use a separate system. Second, reports are slow: staff prepare them by hand, so they are already out of date. Third, there is no early warning: crowding is noticed only after patients complain.

So the hospital has all the data, but not in one place.

Slide 3: The solution (Reflect)

Now the solution. We use three IBM products as one pipeline.

Step one, Collect: IBM DataStage copies data from every hospital system each night. This is called ETL, which means extract, transform and load.

Step two, Store: IBM Db2 Warehouse keeps all that data clean and in one place. This is our data warehouse.

Step three, Analyse: IBM Cognos Analytics reads the warehouse and builds reports and dashboards.

Step four, Decide: managers open one dashboard and take action.

Slide 4: The warehouse design (Reflect)

This is how the warehouse is designed. In the middle is the fact table, called Patient visits. It holds the numbers: waiting time, days stayed and bill amount.

Around it are four dimension tables: Time, Department, Doctor and Patient. They describe each visit.

Because the shape looks like a star, it is called a star schema. It is simple to understand, and Cognos works fast on it.

Slide 5: The build in IBM Cognos (Make)

Now we make it. Inside Cognos there are four steps.

One, Connect: we connect Cognos to the warehouse.

Two, Model: in a tool called Framework Manager, we give the tables simple business names.

Three, Build: we drag and drop waiting time, department and month onto a dashboard. No SQL coding is needed.

Four, Share: we schedule the report, and each department head gets their own figures by email.

Slide 6: The output in IBM Cognos (Make)

This is the kind of dashboard Cognos gives us. These are sample figures for illustration, not real hospital data.

At the top, three tiles show the main numbers: average waiting time, bed occupancy and patient visits.

The bar chart shows that the Emergency department has the longest wait. The line chart shows bed occupancy over six months.

A manager can click on any bar to see more detail.

Slide 7: The result

What changes for the hospital? Before, reports were made by hand, each system showed only part of the picture, and problems were noticed late.

After, the dashboard is ready every morning, all the data is linked, and a crowded department is spotted the same day.

These are the expected benefits of this design.

Slide 8: The IBM values in this solution

This solution also follows IBM's three core values.

First, dedication to every client's success: we started from the hospital's real problem, not from the software.

Second, innovation that matters: the warehouse and Cognos AI are used for something important, which is faster care for patients.

Third, trust and personal responsibility: Cognos uses one governed model with security, so patient data stays correct and protected.

Slide 9: Close

I will end where IBM begins, with one word: THINK.

A data warehouse and Cognos do not think for the hospital. They give its people trusted data and clear answers, so they can think better and give better care.

Thank you. I am happy to take questions.

If the judges ask questions

Question	Short answer
What is a data warehouse?	A central store of historical data from many systems, organised for analysis and reporting.
What is ETL?	Extract, transform, load: taking data out of source systems, cleaning it, and loading it into the warehouse.
What is a star schema?	A design with one fact table of numbers in the middle and dimension tables around it that describe those numbers.
What is IBM Cognos?	IBM's business intelligence tool that turns warehouse data into reports and dashboards.
What is Framework Manager?	The Cognos modelling tool that gives database tables business-friendly names before reports are built.
Is the dashboard real?	No, it is a mock-up with sample figures to show what Cognos would produce.